# Topology for Polynomial Rings

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## The Zero Set of Simultaneous Equations Forms a Topology

### Topological Space

**Topological Space** is a set $X$ with a collection $\mathcal{T} \subseteq 2^X$ is a topological space if $\mathcal{T}$ satisfies the following:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$ (inclusion of the empty set and the entire space).
2. $\mathcal{T}$ is closed under arbitrary unions, i.e., if $\{U_\alpha\}_{\alpha \in A} \subseteq \mathcal{T}$, then $\bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}$.
3. $\mathcal{T}$ is closed under finite intersections, i.e., if $U_1, U_2, \ldots, U_n \in \mathcal{T}$, then $\bigcap_{i=1}^n U_i \in \mathcal{T}$.

**Remark 1.**

- $\{\emptyset, X\} \subseteq \mathcal{T}$
- $\{U_\alpha\}_{\alpha \in A} \subseteq \mathcal{T} \implies \bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}$
- $\{U_i\}_{i=1}^n \subseteq \mathcal{T} \implies \bigcap_{i=1}^n U_i \in \mathcal{T}$

### Zero Set of Polynomials

Let $X$ be a set and let $\mathbb{K}$ be a field (such as $\mathbb{R}$ or $\mathbb{C}$). Consider the ring of polynomials $\mathbb{K}[x_1, x_2, \ldots, x_n]$ in $n$ variables with coefficients in $\mathbb{K}$. For a polynomial $f \in \mathbb{K}[x_1, x_2, \ldots, x_n]$, the zero set of $f : X \to \mathbb{K}$ is defined as:

$$Z(f) = \{\mathbf{x} = (x_1, x_2, \ldots, x_n) \in X \mid f(\mathbf{x}) = 0\} \subseteq X.$$

More generally, for a collection of polynomials $\{f_\alpha\}_{\alpha \in A} \subseteq \mathbb{K}[x_1, x_2, \ldots, x_n]$, the zero set of this collection is defined as:

$$\begin{aligned} Z(\{f_\alpha\}_{\alpha \in A}) &= \left\{\mathbf{x} \in X \mid f_\alpha(\mathbf{x}) = 0 \text{ for all } \alpha \in A\right\} \\ &= \bigcap_{\alpha \in A} \left\{\mathbf{x} \in X \mid f_\alpha(\mathbf{x}) = 0\right\} \\ &= \bigcap_{\alpha \in A} Z(f_\alpha). \end{aligned}$$

**The Zero Set of Simultaneous Equations forms a Topology**

Let
$$\mathcal{T} = \{Z(f) \mid f \in \mathbb{K}[x_1, x_2, \ldots, x_n]\} \subseteq 2^X$$
be the collection of zero sets of polynomials in $\mathbb{K}[x_1, x_2, \ldots, x_n]$. Then $\mathcal{T}$ forms the collection of closed sets for a topology on $X$.

*Proof.* To show that $\mathcal{T}$ forms a topology, we must verify that $\mathcal{T}$ satisfies the following conditions:

1. $X \in \mathcal{T}$ and $\emptyset \in \mathcal{T}$.
2. $\mathcal{T}$ is closed under arbitrary intersections.
3. $\mathcal{T}$ is closed under finite unions.

**1.** $X \in \mathcal{T}$ **and** $\emptyset \in \mathcal{T}$

- $X \in \mathcal{T}$: Consider the constant polynomial $f(\mathbf{x}) = 0$ for all $\mathbf{x} \in X$. The zero set of $f$ is:
$$Z(f) = \{\mathbf{x} \in X \mid f(\mathbf{x}) = 0\} = X.$$
  Thus, $X \in \mathcal{T}$.

- $\emptyset \in \mathcal{T}$: Consider the constant polynomial $g(\mathbf{x}) = 1$ for all $\mathbf{x} \in X$. The zero set of $g$ is:
$$Z(g) = \{\mathbf{x} \in X \mid g(\mathbf{x}) = 0\} = \emptyset.$$
  Thus, $\emptyset \in \mathcal{T}$.

**2.** $\mathcal{T}$ **is Closed Under Arbitrary Intersections**

- Let $\{Z(f_\alpha)\}_{\alpha \in A}$ be a collection of zero sets where $f_\alpha \in \mathbb{K}[x_1, x_2, \ldots, x_n]$ for each $\alpha \in A$.
- Consider the intersection of these zero sets:
$$\bigcap_{\alpha \in A} Z(f_\alpha) = \bigcap_{\alpha \in A} \{\mathbf{x} \in X \mid f_\alpha(\mathbf{x}) = 0\}.$$
- This can be rewritten as:
$$\bigcap_{\alpha \in A} Z(f_\alpha) = \{\mathbf{x} \in X \mid f_\alpha(\mathbf{x}) = 0 \text{ for all } \alpha \in A\}.$$

- Define a new function $g : X \to \mathbb{K}$ by:

$$g(\mathbf{x}) = \sum_{\alpha \in A} f_\alpha(\mathbf{x})^2.$$

  Since $f_\alpha(\mathbf{x})^2 \geq 0$ for each $\alpha \in A$, $g(\mathbf{x}) = 0$ if and only if $f_\alpha(\mathbf{x}) = 0$ for all $\alpha \in A$.

- Therefore, the intersection $\bigcap_{\alpha \in A} Z(f_\alpha)$ is exactly $Z(g)$, which is in $\mathcal{T}$. Thus, $\mathcal{T}$ is closed under arbitrary intersections.

**3. $\mathcal{T}$ is Closed Under Finite Unions**

- Consider two zero sets $Z(f_1)$ and $Z(f_2)$, where $f_1, f_2 \in \mathbb{K}[x_1, x_2, \ldots, x_n]$.
- The union of these zero sets is:

$$Z(f_1) \cup Z(f_2) = \{\mathbf{x} \in X \mid f_1(\mathbf{x}) = 0 \text{ or } f_2(\mathbf{x}) = 0\}.$$

- Define a new function $h : X \to \mathbb{K}$ by:

$$h(\mathbf{x}) = f_1(\mathbf{x}) \cdot f_2(\mathbf{x}).$$

- The zero set $Z(h)$ is given by:

$$Z(h) = \{\mathbf{x} \in X \mid h(\mathbf{x}) = 0\} = \{\mathbf{x} \in X \mid f_1(\mathbf{x}) \cdot f_2(\mathbf{x}) = 0\}.$$

  This is exactly the union $Z(f_1) \cup Z(f_2)$.

- Therefore, $Z(f_1) \cup Z(f_2) = Z(h) \in \mathcal{T}$, showing that $\mathcal{T}$ is closed under finite unions.

$\square$

## Proof that Maximal Ideal $\implies$ Prime Ideal $\implies$ Radical Ideal

### 1. Maximal Ideal $\implies$ Prime Ideal

**Definitions:**

- **Maximal Ideal:** An ideal $M$ in a ring $R$ is **maximal** if:

$$M \subsetneq R \text{ and } \nexists I \text{ such that } M \subsetneq I \subsetneq R.$$

- **Prime Ideal:** An ideal $P$ in a ring $R$ is **prime** if:

$$P \subsetneq R \text{ and } \forall a, b \in R,\ ab \in P \Rightarrow a \in P \text{ or } b \in P.$$

**Proof:** Let $M$ be a maximal ideal in $R$. We need to show that $M$ is a prime ideal.

1. $M$ is proper: Since $M$ is maximal, by definition, $M \subsetneq R$.
2. Prime condition: Suppose $ab \in M$ for some $a, b \in R$. Consider the ideal $(M, a)$ generated by $M$ and $a$:

$$(M, a) = \{m + ra \mid m \in M, r \in R\}.$$

Since $M$ is maximal, there are two possibilities:

$$(M, a) = M \quad \text{or} \quad (M, a) = R.$$

- If $(M, a) = M$, then $a \in M$.
- If $(M, a) = R$, then there exist $m \in M$ and $r \in R$ such that:

$$1 = m + ra.$$

Multiplying both sides by $b$, we get:

$$b = mb + rab.$$

Since $mb \in M$ and $ab \in M$ (because $M$ is an ideal), it follows that $b \in M$.

Hence, $ab \in M$ implies $a \in M$ or $b \in M$, so $M$ is a prime ideal.

$$M \text{ maximal} \implies M \text{ prime}.$$

## 2. Prime Ideal $\Longrightarrow$ Radical Ideal

**Definitions:**

- **Radical Ideal:** An ideal $I$ in a ring $R$ is **radical** if:

$$\forall a \in R,\ a^n \in I \text{ for some } n \geq 1 \Rightarrow a \in I.$$

**Proof:** Let $P$ be a prime ideal in $R$. We need to show that $P$ is also a radical ideal.

1. $P$ is proper: Since $P$ is prime, $P \subsetneq R$.
2. Radical condition: Suppose $a^n \in P$ for some $n \geq 1$. We need to show that $a \in P$.
   - Consider $a^n \in P$. Since $P$ is prime, the definition of a prime ideal implies:

$$\text{If } a^n \in P,\ \text{then } a \in P \text{ or } a^{n-1} \in P.$$

   - If $a \in P$, we are done.
   - If $a^{n-1} \in P$, apply the same argument to $a^{n-1}$:

$$\text{If } a^{n-1} \in P,\ \text{then } a \in P \text{ or } a^{n-2} \in P.$$

   - Repeating this process, we eventually conclude $a \in P$.

Thus, $P$ is a radical ideal.

$$P \text{ prime} \Longrightarrow P \text{ radical}.$$

## Conclusion

By combining the results:

$$\text{Maximal Ideal} \Longrightarrow \text{Prime Ideal} \Longrightarrow \text{Radical Ideal}.$$

This completes the proof.

- $\mathbb{C}[x] = \left\{\sum_{i=0}^{n} a_i x^i : a_i \in \mathbb{C}, n \in \mathbb{Z}_{\geq 0}\right\}$
- $\mathbb{C} = \{\alpha : \alpha \in \mathbb{C}\}$
- Note that $\mathbb{C} \subseteq \mathbb{C}[x]$ since $\mathbb{C} = \{\alpha : \alpha \in \mathbb{C}\} = \left\{0x^n + 0x^{n-1} + \cdots + \alpha x^0 : \alpha \in \mathbb{C}\right\}$
- Consider
$$\begin{array}{rcccl} \phi_\alpha & : & \mathbb{C}[x] & \longrightarrow & \mathbb{C} \\ & & f & \longmapsto & f(\alpha) \end{array}$$
and
- Consider $f(x) \in \mathbb{C}[x]$ s.t.
$$\begin{array}{rcccl} f & : & \mathbb{C} & \longrightarrow & \mathbb{C} \\ & & \alpha & \longmapsto & f(\alpha) \end{array}$$
- By the first isomorphism theorem, we have $\mathbb{C}[x]/\langle x-\alpha\rangle \simeq \mathbb{C}$
- $\langle x-\alpha\rangle = \left\{(x-\alpha)f(x) : f(x) \in \mathbb{C}[x]\right\} \subseteq \mathbb{C}[x]$
- $\mathbb{C}[x]/\langle x-\alpha\rangle$;
  - $p(x) = (x-\alpha)q(x) + r(x) = (x-\alpha)q(x) + r$
  - $\mathbb{C}[x]/\langle x-\alpha\rangle = \left\{r + \langle x-\alpha\rangle : r \in \mathbb{C}\right\}$

| **Mapping** | $\begin{array}{rcccl} \psi_p & : & \mathbb{Z} & \longrightarrow & \mathbb{Z}_p \\ & & n & \longmapsto & n \bmod p = \psi_p(n) \end{array}$ | $\begin{array}{rcccl} \phi_a & : & \mathbb{C}[x] & \longrightarrow & \mathbb{C} \\ & & f(x) & \longmapsto & f(a) = \phi_a(f(x)) \end{array}$ |
|---|---|---|
| Additive Homo. | $\psi_p(a+b) := (a+b) \bmod p$ | $\phi_a(f+g) := f(a) + g(a)$ |
| Multiplicative Homo. | $\psi_p(ab) := (ab) \bmod p$ | $\phi_a(fg) := f(a)g(a)$ |
| Kernel | $\ker(\psi_p) = p\mathbb{Z}$ | $\ker(\phi_a) = (x-a)\mathbb{C}[x]$ |
| Image | $\mathbb{Z}_p$ | $\mathbb{C}$ |
| Ideal | $p\mathbb{Z} = \langle p\rangle$ | $(x-a)\mathbb{C}[x] = \langle x-a\rangle$ |
| Prime Ideal | $\langle p\rangle$ is prime | $\langle x-a\rangle$ is prime |
| Maximal Ideal | $\langle p\rangle$ is maximal | $\langle x-a\rangle$ is maximal |
| Isomorphism | $\mathbb{Z}_p \simeq \mathbb{Z}/p\mathbb{Z}$ | $\mathbb{C} \simeq \mathbb{C}[x]/\langle x-a\rangle$ |
| Element of Domain | $\begin{array}{rcccl} n & : & arg2 & \longrightarrow & arg3 \\ & & arg4 & \longmapsto & arg5 \end{array}$ | $\begin{array}{rcccl} f & : & \{\langle x-\alpha\rangle : \alpha \in \mathbb{C}\} & \longrightarrow & \coprod_\alpha \mathbb{C}[x]/\langle x-\alpha\rangle \\ & & \langle x-\alpha\rangle & \longmapsto & arg5 \end{array}$ |

| | | 2 | 3 | 5 | 7 |
|---|---|---|---|---|---|
| $\varphi(2\mathbb{Z})$ | $\longrightarrow 2\mathbb{Z}$ | 1 | 0 | 0 | 0 |
| $\varphi(3\mathbb{Z})$ | $\longrightarrow 3\mathbb{Z}$ | 0 | 1 | 0 | 0 |
| $\varphi(5\mathbb{Z})$ | $\longrightarrow 5\mathbb{Z}$ | 0 | 0 | 1 | 0 |

Prime numbers $p$